Practice2_Barraza_Torres

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Objectives: Understand the use of *****tables**; **deviation**, histograms and bar **graphs**; and the **descriptive statistics** on a table relied on another file

1. Importation of the data on an **excel file**

```
library(readxl)
data <- read_excel("C:/Users/USER/Downloads/DatosDeLaEncuesta.xlsx") #Import

View(data)  #This one is for verification
```

2. Visualization of the data on a **table**

```
nrow(data) #Rows
```

```
## [1] 52
```

```
ncol(data) #Columns
```

```
## [1] 2
```

```
str(data) #Summary
```

```
## tibble [52 x 2] (S3: tbl_df/tbl/data.frame)
##  $ Edad_Trabajo_estable: num [1:52] 24 35 52 21 35 24 25 25 25 20 ...
##  $ Carrera             : chr [1:52] "TI" "TI" "TI" "ENRE" ...
```

```
head(data) #First registers
```

```
## # A tibble: 6 x 2
##   Edad_Trabajo_estable Carrera
##                  <dbl> <chr>
## 1                   24 TI
## 2                   35 TI
## 3                   52 TI
## 4                   21 ENRE
## 5                   35 TI
## 6                   24 DN
```

```
tail(data) #Last registers
```

```
## # A tibble: 6 x 2
##   Edad_Trabajo_estable Carrera
##                  <dbl> <chr>
## 1                   41 ENRE
## 2                   18 ENRE
## 3                   47 OCI
## 4                   65 TI
## 5                   78 TI
## 6                   63 DISEÑO
```

3. Assignation of **rows** and **columns**

```
age <- data$Edad_Trabajo_estable
carreer <- data$Carrera
```

4. Import of data **frequencies**

```
freq_carrer <- table(carreer)
print(freq_carrer)
```

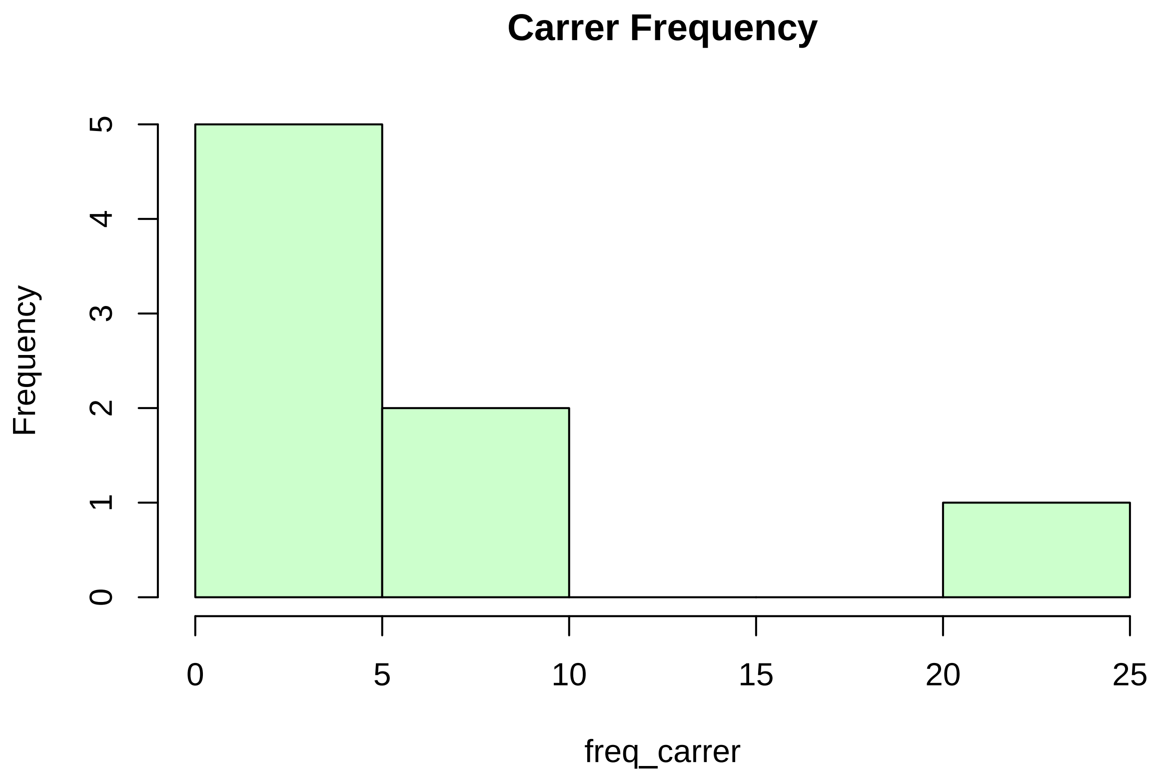
```
## carreer
##  DISEÑO      DN    ENRE L. ING.    MECA      MI     OCI      TI
##       5       6       5       3       4       1       6      22
```

```
freq_age <- table(age)
print(freq_age)  #This one is rare to use
```

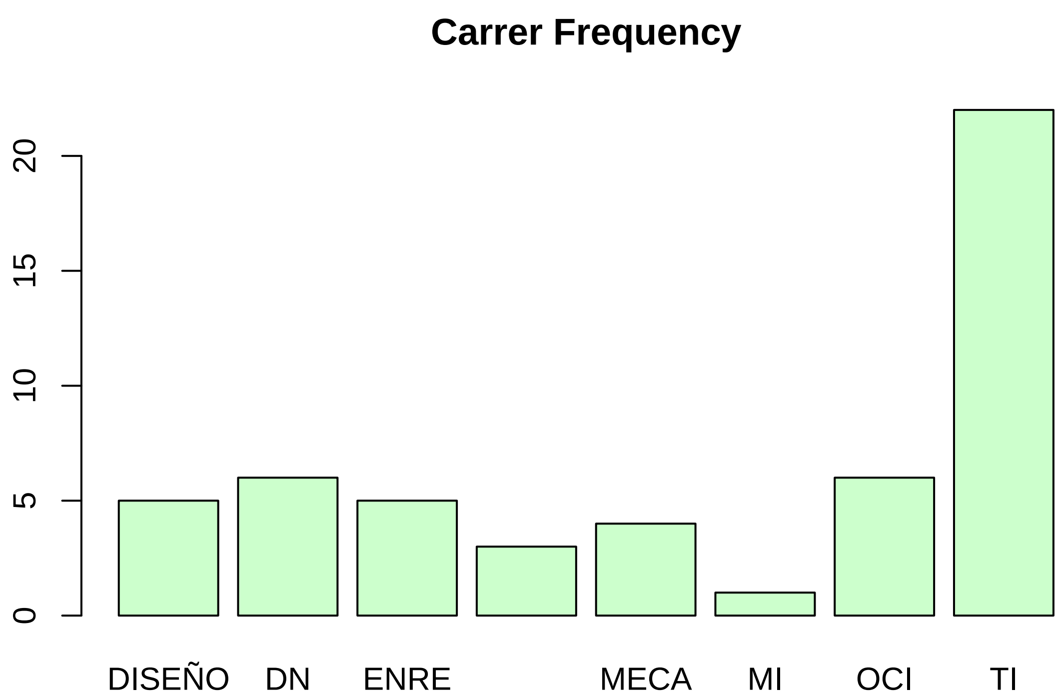
```
## age
## 18 20 21 22 23 24 25 26 27 28 29 30 32 33 35 36 41 42 47 52 56 60 62 63 65 78
##  1  1  1  1  1  6  4  2  1  3  3  1  2  1  3  3  4  1  1  3  1  1  1  2  1  1
## 80 98
##  1  1
```

5. Type of **graphics**: Bar graph, histogram and frequency graph

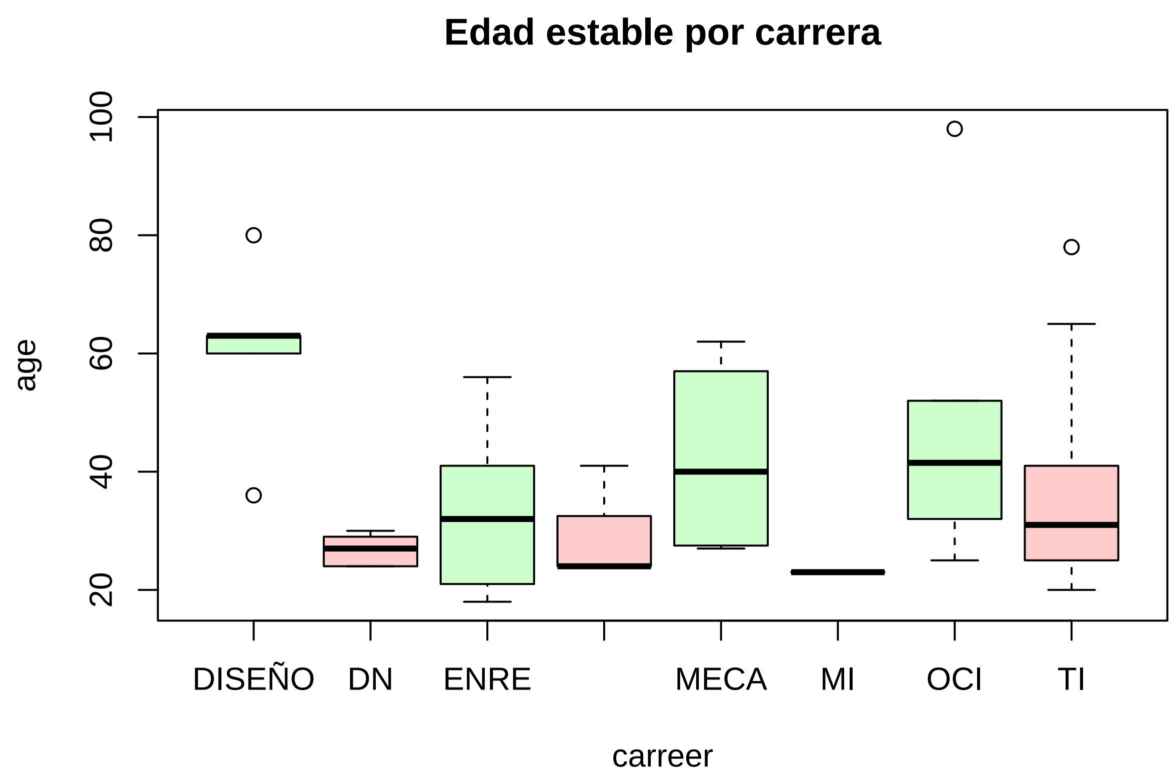
```
#Histogram
hist(freq_carrer, col="#CCFFCC", main = "Carrer Frequency")
```




```r
#Bar graph
barplot(freq_carrer, col="#CCFFCC", main = "Carrer Frequency")
```




```r
#Frequency graph
boxplot(age ~ carreer, col = c("#CCFFCC", "#FFCCCC"), main= "Edad estable por carrera")
```



6. **Descriptive statistics**

```
mean(freq_carrer) #Mean
```

```
## [1] 6.5
```

```
median(freq_carrer) #Median
```

```
## [1] 5
```

```
mode <- table(carreer)
print(mode) #Mode
```

```
## carreer
##   DISEÑO       DN     ENRE L. ING.     MECA       MI      OCI       TI
##        5        6        5        3        4        1        6       22
```

```
range(freq_carrer) #Range
```

```
## [1]  1 22
```

```r
var(freq_carrer) #Variance
```

```
## [1] 42
```

```r
sd(freq_carrer) #Standart desviation
```

```
## [1] 6.480741
```

Thanks for watching